



Micro UAV Autonomous Navigation System With Deep Learning based Monocular Vision Depth Estimation

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Introduction

Background

- Understanding the three-dimensional characteristics of the flight space is important for drone obstacle avoidance. Research on environmental perception using LiDAR, stereo cameras, and monocular cameras is actively underway.
- Small drones have limited payload capacity and battery capacity, which limits the number of sensors they can use. Research addresses these constraints by using small sensors and remote computers.
- Drones are increasingly used for indoor exploration in locations that are difficult for people to access.

Research Objectives

Environmental Perception for Small Drones

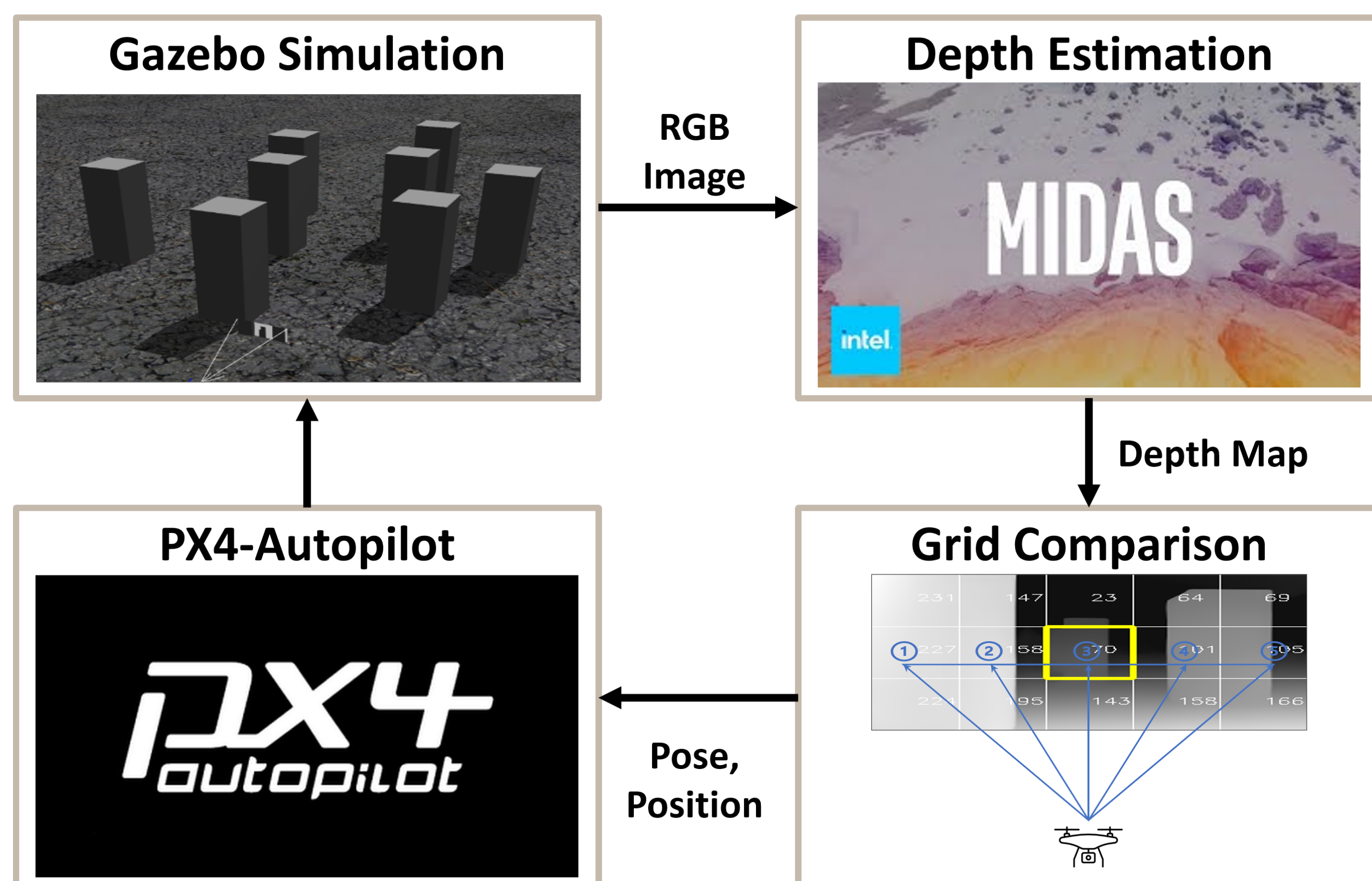
- Use a deep-learning-based monocular camera depth-estimation model to estimate relative distances ahead of the flight region.
- Use depth information to estimate obstacle density and identify a safe region ahead.

Development of Drone Obstacle Avoidance

Develop a drone-control system that determines the flight direction from depth information, periodically updates the path, and flies in the selected direction.

Methods

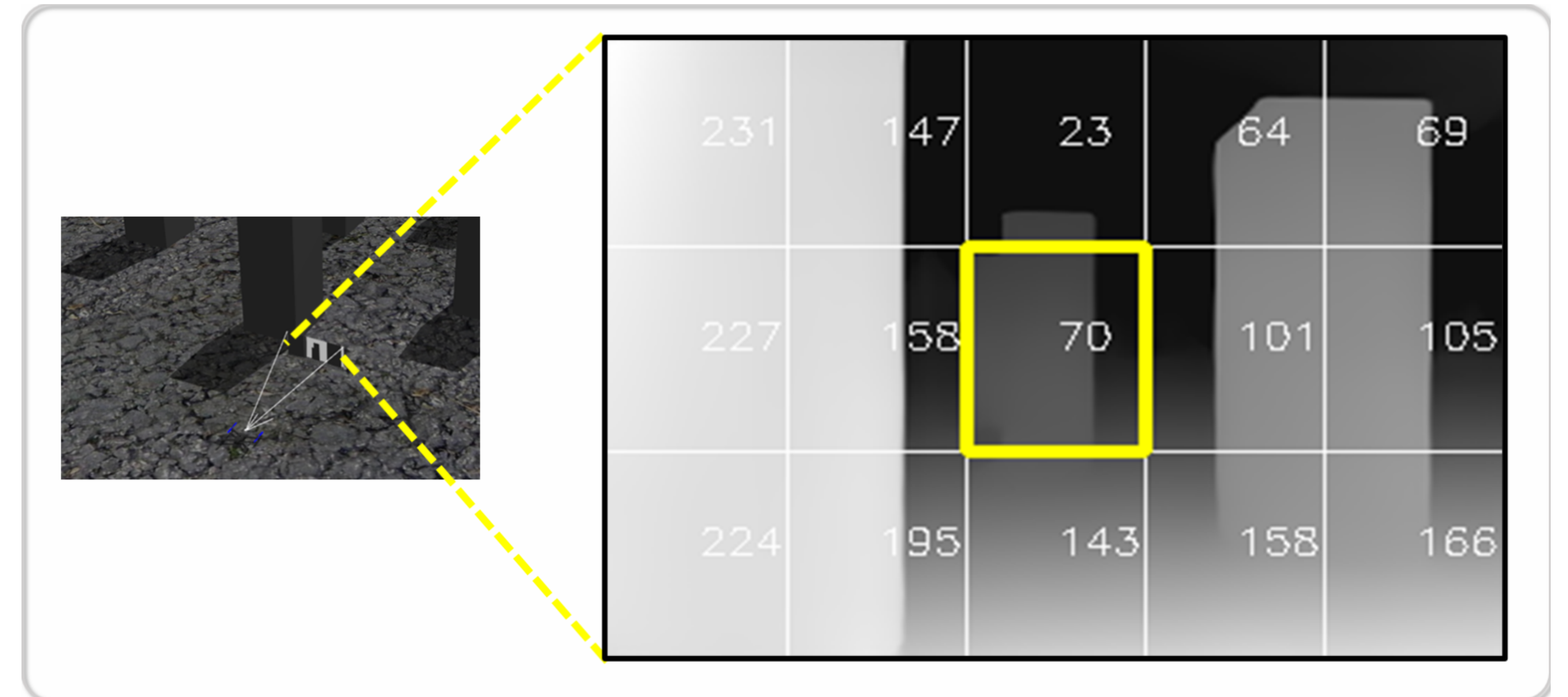
System Overview



System overview: RGB image, depth map, pose/position, and the five candidate flight directions.

- Build a Gazebo simulation environment to test the flight algorithm, using the open-source PX4-Autopilot package for the drone.
- Use the MiDaS deep-learning model for monocular camera depth estimation. A single RGB image is the input; the model estimates relative distances between pixels and expresses them as a depth map with values from 0 to 255.
- Develop the system using ROS to exchange image-processing data and flight commands.

Flight Direction Decision

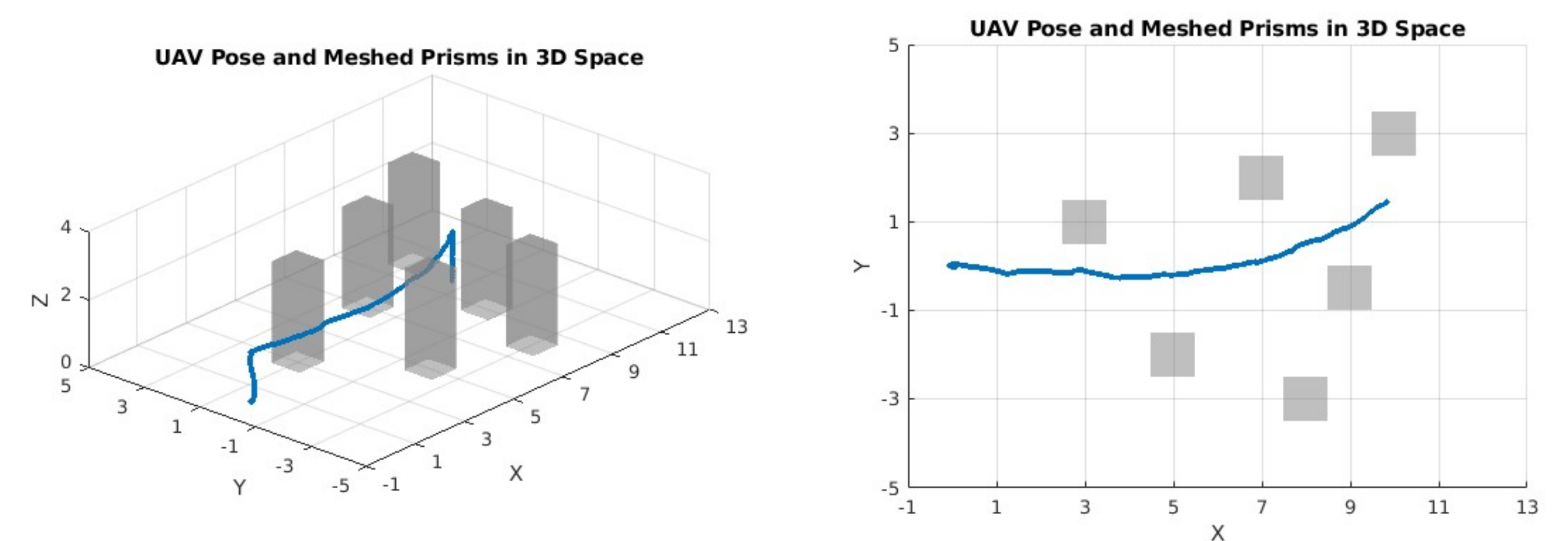


Obstacle scene and per-cell mean depth values; yellow indicates the selected cell.

- Divide the depth-information image generated by MiDaS into a 5×3 grid. Calculate the mean depth of each cell to estimate obstacle density.
- Rotate the drone's yaw toward the cell with the lowest mean value, then move forward in that direction. Repeat path updates in 0.5 m steps.

Result

Test Result



3D and top-view trajectories. The blue line shows the simulated flight path.

- Test the algorithm in a flight environment with six rectangular pillars randomly placed within 10 m ahead of the drone's starting position.
- Confirm that image processing from a single RGB image to a depth map can run at 41 FPS.
- In the simulated flight experiment, the drone takes off to a height of 1.5 m and completes a 10 m forward flight without colliding with the obstacles.

Conclusion

- Develop an obstacle-avoidance technique for small drones using a monocular camera depth-estimation model.
- Apply the deep-learning model for monocular camera depth estimation to a ROS-based system and confirm that it can support obstacle avoidance in real time.
- Test the ROS-based obstacle-avoidance algorithm in simulation and confirm that forward flight is possible while avoiding randomly placed obstacles ahead.
- In future work, add altitude control to avoid obstacles of various shapes and evaluate performance through field tests with an actual drone.